



Building back greener: Promoting performance in natural resource markets for sustainable development

Xi Chen^a, Binyi Liu^{a,*}, Yali Ma^b, Shuhui Lv^c

^a College of Architecture and Urban Planning, Tongji University, Shanghai, 200092, China

^b Regional Development Research Department, Zhejiang Provincial Development Planning Research Institute, Hangzhou, 310030, China

^c Department of Management Engineering, Shanxi Youth Vocational College, Taiyuan, 030032, China

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ABSTRACT

The problem of unsustainable development of natural resource markets is a critical challenge to the country's long-term prosperity and global ecological health. China has experienced unprecedented economic growth, but this has come at significant environmental costs. Using a Panel Data model, this study aims to investigate the effectiveness of various policy interventions aimed at "Building Back Greener" in China's natural resource markets over the period 2000–2020. The research focuses on assessing the impact of regulatory measures, market incentives, and public-private partnerships on promoting sustainable practices. The results indicate that a multi-pronged approach, combining strict regulations with market-based incentives, is most promising in enhancing performance and sustainability in natural resource markets. These findings offer valuable insights for policy-makers, industry stakeholders, and environmental advocates, highlighting the need for a balanced approach to achieve sustainable development in China and potentially other rapidly developing economies.

1. Introduction

The urgent need to solve the serious environmental problems facing our world drives the case for improving natural resource market efficiency for sustainable development. There must be a paradigm shift in favor of more sustainable activities due to climate change, biodiversity loss, pollution, and resource depletion (Lu et al., 2023). Policymakers and stakeholders need to improve the efficiency of natural resource markets due to rising awareness of consumers, investors, and civil society and their demands for sustainability, which is on line with social expectations and values (X. F. Liu et al., 2023).

There is a growing awareness of the urgent need to "build back greener" for sustainable growth in the face of worsening environmental problems and global disasters. To reduce the adverse effects of human activities on the environment, this idea highlights the need to move towards more ecologically sustainable behaviors and policies. Increasing the natural resource market efficiency is an integral part of this transformation (Tong et al., 2023). Achieving sustainability objectives and maintaining long-term economic growth depend on the wise allocation and use of natural resources. Depletion and improper management of natural resources seriously threaten the environment and human well-being. Energy generation is just one business that depends

on these resources, including water, minerals, forests, and energy. However, waste, environmental deterioration, and social inequities are the result of inefficient resource allocation and usage methods. Promoting sustainable development and achieving a healthy balance between economic growth, social welfare, and environmental protection depends on correcting inefficiencies in natural resource markets (Guo et al., 2023).

Given the enormous environmental concerns and their interrelated effects on economies and communities, there is an increasing need to shift to sustainable practices. The notion of "constructing a more environmentally friendly future" has developed as a compelling slogan for synchronizing post-crisis recovery efforts with the principles of ecological preservation and optimal resource use. This research explores the complex and diverse field of natural resource management, examining several approaches and policy measures designed to improve efficiency and long-term sustainability in this key area. Natural resources play a fundamental role in the foundations of economies, serving as the basis for industries, livelihoods, and general human well-being. However, the use of these resources often leads to adverse consequences such as ecological degradation, depletion of natural resources, and worsening of climate change (Lin et al., 2023). The need to deal with these difficulties while promoting economic

* Corresponding author.

E-mail address: byliutju@sina.com (B. Liu).

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expansion requires a nuanced equilibrium. In light of recent global disasters, such as the COVID-19 epidemic, governments around the globe are presented with a distinct and advantageous moment to reconfigure their resource management approaches (Ullah et al., 2020; Fu et al., 2023). Comprehensive natural resource management encompasses more than just preservation; it requires a holistic perspective that takes into account environmental, social, and economic aspects. The transition to sustainable practices requires a comprehensive reassessment of extraction methodologies, enhancement of resource use efficiency, and incorporation of circular economy ideas. According to (Fang et al., 2022), policymakers can achieve the dual goals of mitigating resource depletion and promoting resilience and social equality by adopting a systems thinking approach. Achieving sustainable resource management requires the use of a range of new policy tools that effectively align economic incentives with the goal of environmental preservation. Carbon pricing techniques, such as the internalization of resource exploitation externalities through carbon pricing, redirect investment towards cleaner technologies and practices. Similarly, implementing eco-labeling and certification initiatives enables customers to make well-informed decisions, motivating companies to embrace sustainable practices to maintain a competitive edge (Li and Umair, 2023). Integrating sustainable financing and investment practices is a crucial component in the effort to construct a more environmentally friendly future. Growing awareness among financial institutions of the potential hazards associated with resource depletion and environmental degradation has led to a notable increase in market demand for environmentally sustainable financial solutions. According to (Wu et al., 2022), investors now tend to allocate their financial resources to ecologically sustainable projects. This shift in investment patterns not only promotes innovation but also facilitates the adoption of resource-efficient technology.

Technological innovation plays a key role in facilitating resource performance improvements. Advances in sustainable energy, innovative materials, and digital technologies show the potential to decouple economic expansion from the use of finite resources. As an example, the use of renewable energy technologies has been shown to reduce the carbon intensity associated with resource extraction. Additionally, advances in materials science have led to improved product longevity, reducing the need for frequent replacements (F. X. Liu et al., 2023). The importance of global cooperation cannot be overstated in the context of a highly integrated global landscape characterized by economic and environmental complexities. Facilitating a coordinated transition towards greener resource management can be achieved through the exchange of best practices, the transfer of sustainable technology, and policy alignment. The promotion of collaboration and the equitable advancement of sustainable development are facilitated by the inclusion of multilateral platforms and agreements, which aim to prevent any country from being excluded (Pan et al., 2023).

The primary contribution of this study lies in its comprehensive approach to understanding the dynamics of natural resource markets in China over the time period 2000–2020. Unlike previous studies that either focused solely on regulatory measures or market-based incentives, this research uses a Panel Data model to capture the longitudinal effects of multiple policy interventions. In doing so, it offers a more nuanced understanding of what works and what doesn't in the pursuit of sustainable development. Our results show that a balanced, multifaceted approach that combines stringent regulations with market-based incentives is most effective in promoting sustainability in natural resource markets. Specifically, the study finds that regulatory measures alone are insufficient and need to be complemented by market incentives to encourage sustainable practices among producers and consumers. These findings have significant implications for policymakers, industry stakeholders, and environmental advocates, not just in China, but also in other rapidly developing economies that face similar challenges. By integrating a time-sensitive model and focusing on a critical period of economic and environmental change, this study provides practical

insights that can help China, and potentially other countries, in “Building Back Greener” for a more sustainable future.

2. Literature review

In recent years, there has been a substantial increase in interest in the role of artificial intelligence (AI) in creating a green economy. AI has enormous potential to improve green total factor productivity (TFP) and natural resource management as civilizations around the world work to solve environmental problems and transition to more sustainable practices (Chen et al., 2023). This literature review focuses on AI's role in fostering sustainable development and accelerating economic growth in an ecologically responsible manner. It also examines the combined drivers of natural resources and green TFP. Natural resources, including water, minerals, forests, and energy, are essential for economic growth. However, their extraction and use often have a significant negative impact on the environment. Resource depletion, pollution, and environmental degradation are problems caused by ineffective management and distribution of resources. Resource consumption must be optimized, waste must be reduced, and environmental impacts must be minimized to achieve sustainable development (Zhang et al., 2020). Green TFP examines how effectively inputs, such as natural resources, are converted into desired outputs, while considering environmental impacts. It encapsulates the idea of “doing more with less” and illustrates how an economy can provide economic development while using few resources and causing little damage to the environment. To promote economic growth without sacrificing environmental sustainability, improving green TFP is a key component of sustainable development (Yan et al., 2021).

The technology that enables robots to perform cognitive functions that traditionally require human intellect is called artificial intelligence. By maximizing resource allocation, enhancing resource efficiency, and assisting decision-making processes, AI has enormous potential to promote sustainable development (Yin and Song, 2023). AI-powered applications, such as data analytics, machine learning, and optimization algorithms, can support sustainable growth in several different fields. By examining large data sets, finding patterns, and forecasting future trends, AI systems may aid in efficiently using natural resources. AI algorithms can improve irrigation systems in agriculture, ensuring efficient water utilization and reducing wastage. With the help of AI, energy resources can be managed wisely, with distribution and consumption patterns optimized to reduce carbon emissions (Yi et al., 2023). AI can also help with sustainable forestry practices by identifying regions vulnerable to deforestation and implementing preventative measures. AI contributes significantly to green TFP by driving innovation and efficiency. Machine learning algorithms can analyze large data sets to find areas where there is potential to reduce energy, emission, and waste. Industries can optimize resource utilization, reduce production waste, and boost energy efficiency by incorporating AI into industrial processes. In addition, AI-driven optimization models help urban planning by creating sustainable transportation and infrastructure.

Despite the potential advantages, there are still barriers to AI adoption in green economic growth. Ethical issues such as data privacy, algorithm bias, and transparency must be addressed to promote the ethical and fair adoption of AI. To fully use AI in sustainable development, there is also a need for capacity building and skills development. Cooperation between governments, businesses, academia, and civil society is critical to creating legal frameworks, improving information sharing, and supporting the ethical AI adoption (Lang et al., 2016). Numerous case studies and practical investigations have shown how useful AI is for the growth of the green economy. For instance, using AI-driven energy management systems, energy prices and usage in commercial buildings have been dramatically reduced. AI-powered smart grids have enabled the efficient integration of renewable energy sources into current energy networks. Additionally, the use of AI in waste management, transportation, and disaster response has shown promise for reducing

environmental impacts and boosting resilience (H. Liu et al., 2021).

The use of natural resources has emerged as a prominent subject of concern in contemporary times, unlike in previous decades. (Xiuzhen et al., 2022) studied the indices of the natural resource consumption efficiency. The authors argue that environmental impact assessment can serve as a significant indicator of the efficiency of natural resource consumption. A study by (Mohsin, Zhou, et al., 2018) investigates the impact of advances in green innovation on the efficiency of natural resource use. The first findings support the positive impact of sustainability strategies on the efficiency of natural resource use. (Mohsin et al., 2019) conducted a study to examine the efficacy with which water resources are used in poor nations. The results of the panel Tobit model support the idea that improved use of water resources in the agricultural economy and the adoption of more efficient irrigation practices have positive benefits. (Mohsin, Taghizadeh-Hesary, et al., 2021) conducted a study using supply chain analysis to assess the efficiency of resource use in developing nations in the period 1994–2019. The researchers find that consumption efficiency is comparatively low across nations and suggest the implementation of new regulatory measures to enhance it. (Mohsin, Rasheed, et al., 2018) argue that the effectiveness of a nation's natural resource management depends on the proficiency with which its population harnesses and utilizes these resources. Therefore, there is a need to enhance the efficient use of natural resources in order to guarantee sufficient food supply for present and future generations. (Mohsin, Hanif, et al., 2021) used a pooled mean group model to assess the efficiency of water usage in underdeveloped nations. The findings indicate a lack of efficient use of water resources in developing nations, highlighting the need for effective interventions to improve the current state of affairs. The study conducted by investigates the relationship between urbanization and natural resource use efficiency (NRUE) in emerging urban areas. The findings show a significant association between NRUE and many indicators, including the general population, the economy, and per capita expenditures. The previous study mainly concentrated on examining the correlation between NRUE and reduction of greenhouse gas emissions, with the intention of improving the understanding of the interrelationship between sustainable development practices and preservation of natural resources. Empirical models show a statistically significant relationship between these two elements. Enhancing the efficiency of natural resource use through various measures has the potential to enhance green, sustainable prosperity. Based on research conducted by (Xia et al., 2020), it has been suggested that a country's population can experience growth when technical progress contributes to a better use of its natural resources. Consequently, legislators should pay more attention to technological innovation. The objective of the parameter estimate methodology by (Mohsin et al., 2020) is to determine the impact of NRUE on enhancing the sustainability of the green economy in developing nations. The main results indicate that NRUE has a positive impact on environmentally sustainable economic activities through reducing energy consumption and increasing energy efficiency. The study by investigates the relationship between resource efficiency and the use of renewable energy, with a particular emphasis on sustainability in economic growth. Effective use of resources has been shown to have a positive correlation with ecological advancement. This study examines the correlation between waste recycling practices and the promotion of sustainable environmental development. The paper suggests that the use of advanced technologies can potentially enhance operational efficiency and mitigate energy consumption in the context of technological practices. Consequently, a country has the potential to enhance the efficiency of its use of natural resources by strengthening its green innovations. The use of natural resources plays a key role in promoting sustainable development. Consequently, governments have the potential to enhance their environmentally conscious economic growth by optimizing the efficiency of their current natural resource use practices. Research by (Asbahi et al., 2019) demonstrates that green development in China can be enhanced by implementing more efficient strategies for the use of natural resources, with a particular focus on

coal. Tenant mismanagement is considered one of the sociocultural causes that contribute to underdevelopment. The provision of large rents can potentially catalyze the involvement of corrupt governmental authorities in embezzlement activities, diverting funds intended for public utilization. This illicit behavior has the potential to undermine the provision of essential public services and poses a significant threat to the nation's overall security and stability. The engagement of political actors in organizational affairs has the potential to result in the development of inefficient administrative structures characterized by chronic financial deficits. Moreover, empirical evidence shows that transitory surges in commodity income can potentially undermine the democratic legitimacy of governments. However, even when considering the wholesale level, the available knowledge remains inadequate to draw definitive conclusions about the extent to which organizations contribute to the aggravation of the tragedy of the commons. Property rentals not only represent a significant social issue but also serve as a catalyst for unethical behavior. Based on recent scholarly research conducted by, it has been observed that there is a lack of evidence to support the existence of the tragedy of the commons or its positive impact on economic growth. Countries with abundant natural resources can either experience limited growth or achieve significant expansion, depending on the extent of financial resources. Research shows that even prominent entities such as international banks are subject to competitive pressures influenced by established sovereignty and bureaucratic structures, which shape their development over time. (Ikram et al., 2019) argue that the need to examine empirical evidence and incorporate national and sector-specific regression models is underscored by the ambiguity surrounding this issue, as well as the tendency of government entities to allocate funds, while numerous emerging governments opt for corporate entities.

3. Study design

3.1. Model setting

The impact of sustainable development (SD) on eco-friendly economic growth is studied using a static panel model that relates SD to green total factor productivity. The explained variable is green total factor productivity, the major explanatory variable is SD, and the control variables are other elements that affect the building back greener policy (Khatrri and Singh, 2022). The specific layout is as follows:

$$BBG_{it} = \alpha + \gamma \times NR_{it} + \sum_j \eta_j X_{jit} + \mu_i + \nu_t + \varepsilon_{it} \quad (1)$$

Based on the findings of the aforementioned theoretical study that SD has a nonlinear U-shaped impact on building back greener, we construct a nonlinear regression model that incorporates a squared term for the degree of AI. The nonlinear static panel regression model is represented by Equation (Eq) (McKeen and Fung, 2014).

$$BBG_{it} = \alpha + \gamma_1 \times NR_{it} + \gamma_2 \times NR_{it}^2 + \sum_j \eta_j X_{jit} + \mu_i + \nu_t + \varepsilon_{it} \quad (2)$$

$$BBG_{it} = \alpha + \beta \times BBG_{i,t-1} + \gamma_1 \times NR_{it} + \gamma_2 \times NR_{it}^2 + \sum_j \eta_j X_{jit} + \varepsilon_{it} \quad (3)$$

Statisticians use the extended technique of moments to compute and assess dynamic panel models. Both the generalized difference method of moments and the GSM of moments fall under the umbrella of the GMM. In the GMM of moments (MacNaughton et al., 2018), the difference equation is computed using a first-order difference processing using lag variables. The generalized system method of moments enhances estimation efficiency over the difference generalized method of moments by combining difference and level equations. We use two estimation methodologies to guarantee the reliability of the estimated results, while the gap between the generalized method of moments and the generalized system method of moments fluctuates (Islam et al., 2015).

$$P^x(x) = \begin{cases} (y^t, b^t) : \sum_{k=1}^K \beta_k^t y_{km}^t \geq y_{km}^t, \forall m; \sum_{k=1}^K \beta_k^t b_{ki}^t = b_{ki}^t, \forall i; \\ \sum_{k=1}^K \beta_k^t x_{kn}^t \leq x_{kn}^t, \forall n; \sum_{k=1}^K \beta_k^t = 1, \beta_k^t \geq 0, \forall k \end{cases} \quad (4)$$

$$P^G(x) = \begin{cases} (y^t, b^t) : \sum_{t=1}^T \sum_{k=1}^K \beta_k^t y_{km}^t \geq y_{km}^t, \forall m; \sum_{t=1}^T \sum_{k=1}^K \beta_k^t b_{ki}^t = b_{ki}^t, \forall i; \\ \sum_{t=1}^T \sum_{k=1}^K \beta_k^t x_{kn}^t \leq x_{kn}^t, \forall n; \sum_{t=1}^T \sum_{k=1}^K \beta_k^t = 1, \beta_k^t \geq 0, \forall k \end{cases} \quad (5)$$

$$\text{s.t. } \sum_{t=1}^T \sum_{k=1}^K \beta_k^t y_{km}^t - s_m^y = y_{km}^t, \forall m; \sum_{t=1}^T \sum_{k=1}^K \beta_k^t b_{ki}^t + s_i^b = b_{ki}^t, \forall i; \\ \sum_{t=1}^T \sum_{k=1}^K \beta_k^t x_{kn}^t + s_n^x = x_{kn}^t, \forall n; \sum_{t=1}^T \sum_{k=1}^K \beta_k^t = 1; \beta_k^t \geq 0, \forall k \quad (6)$$

$$GML_t^{t+1} = \frac{1 + \overrightarrow{D}_v^G(x^t, y^t, b^t, g^x, g^y, g^b)}{1 + \overrightarrow{D}_v^G(x^{t+1}, y^{t+1}, b^{t+1}, g^x, g^y, g^b)} \quad (7)$$

Green technological progress (GTC) is an umbrella term that covers developments in sustainable production and associated areas. Green Effectiveness Development, or GEC for short, focuses on more efficient allocation of resources and operations management. A specific decomposition is shown (Feng et al., 2019):

$$GTC_t^{t+1} = \frac{[1 + \overrightarrow{D}_v^G(x^t, y^t, b^t, g^x, g^y, g^b)] / [1 + \overrightarrow{D}_v^G(x^t, y^t, b^t, g^x, g^y, g^b)]}{[1 + \overrightarrow{D}_v^G(x^{t+1}, y^{t+1}, b^{t+1}, g^x, g^y, g^b)] / [1 + \overrightarrow{D}_v^{t+1}(x^{t+1}, y^{t+1}, b^{t+1}, g^x, g^y, g^b)]} \quad (8)$$

$$GEC_t^{t+1} = \frac{1 + \overrightarrow{D}_v^G(x^t, y^t, b^t, g^x, g^y, g^b)}{1 + \overrightarrow{D}_v^G(x^{t+1}, y^{t+1}, b^{t+1}, g^x, g^y, g^b)} \quad (9)$$

$$BBG_{t+1} = GML_t^{t+1} \times BBG_t \quad (10)$$

3.2. Data sources

Industries with commonalities between the International Financial Reporting Standards and the China Industrial Statistics Classification, which use different divisions and names for their sub-sectors, need comparison and classification. This study integrates two sets of criteria for classifying things by analyzing panel data on provinces in China from 2000 to 2020. This data spans 2005–2018 and comes from the International Robot Alliance Website, the China Statistical Yearbook, and the China Science and Technology Statistical Yearbook.

3.3. Control variables

SD and the natural resource market influence the need for labor in the green economy. Large industrial businesses that employ many people and consume a lot of energy tend to have significant greenhouse gas emission intensities (Wut, 2020). This is even more true for the manufacturing sector. However, high-tech companies with few workers have low carbon intensity. Therefore, we tallied up the sector's workforce when the year ended.

Income has both a positive and negative impact on CO2 emissions. First, when incomes rise, firms have more incentive to invest in machinery and tools that can increase output. Rising wealth will encourage

businesses to expand their reliance on fossil fuels, leading to higher levels of CO2 in the atmosphere. Increasing wages and using better industrial equipment can reduce emissions. We divided the annual revenues by the total number of employees in the sector to understand the sector's prosperity. Industry research and development budgets have been used to quantify technological progress. Foreign direct investment (FDI) is made. Foreign direct investment (FDI) can enhance environmental standards and technologies by investing in the host country and spreading new knowledge. This will help the host country's environmental preservation efforts (Osei-Kyei et al., 2021). However, these companies often import highly polluting businesses into the host country using the laxer environmental regulatory framework common in developing countries. FDI was measured in terms of foreign ownership stakes (Bertone et al., 2018).

Energy price (EP). Demand in the volatile sector shifts as fossil fuels give way to renewables as energy costs rise. The need to enhance and optimize energy efficiency in response to this has grown. This study used the China Coal Price Index to observe changes in energy prices. We used descriptive statistics to analyze the most critical variables. The model was found to be resistant to multicollinearity since the majority of correlation values between the independent variables were less than or equal to 0.5215. Detailing the variance expansion components, we found that all of them were less than the empirical threshold of 10. Therefore, the model has no serious multicollinearity problems.

4. Empirical results and discussion

The idea that a product's worth should be determined by the number of hours of effort put into it has been further undermined by the proliferation of free downloadable products and services available on the Internet. Marketing is used to earn money from free Internet services, including searches, maps, emails, cutting-edge phone apps, and movies. For example, a subscription-based pricing model can be used for online video games, while portable gaming equipment can be discounted to encourage software game developers. Their intended purpose is to increase traffic to business websites and to increase product acceptance through networking. This has nothing to do with the value of human work; instead, **sustainable development** is one of the most significant technological advances of the last several decades. Sustainable development has the potential to drastically change the way people experience and interact with the real and virtual worlds, with repercussions for their thoughts, feelings, and deeds at local, national, and international scales. Sustainable development has enormous potential, which has resulted in fierce rivalry among countries, businesses, and individuals to achieve the highest levels of success in this field.

According to new research from the World Economic Forum, SD is already being used in many sectors, making it a fundamental invention for governments and corporations. As intelligent systems continually improve the way hazardous waste, electronic waste, radioactive materials, and other types of garbage are collected, distributed, classified, and recovered, so are the practices related to other types of pollution. (Poirazis et al., 2008) wants to squander. SD can potentially improve Britain's and other nations' economies and global status. How people interact with and react to SD will determine its trajectory in history (Zhao and Li, 2022). Using SD might help solve several social problems. An ageing population, the ongoing climate problem, and the need for rescue efforts fall into this category. Future social-ecological problems

will require more human intelligence, but SD is leading the way (Lam et al., 2010). By learning through interactions with people and the world, these systems will help people make decisions and assess opportunities. If developed correctly, SD might impact society in a way comparable to the Industrial Revolution. This study focuses on AI's applications in SWM and examines the potential benefits, opportunities, and ramifications of these ground-breaking new methods.

Machines and computers are considered intelligent if they mimic human mental processes such as learning and error detection (Huberman and Pearlmutter, 2008). A natural resource with human-like perception and cognition is the result of several methods, computing approaches, and tactics. SD is a branch of computer science that analyzes the environment and adjusts its actions appropriately to increase the likelihood of success. AI can correct errors, make sound choices, and respond swiftly to changing circumstances.

4.1. LSDVC method analysis

There is consensus among experts that SD consists of four main parts. Heuristic conflict resolution uses educated guesses and a limited set of options to reach satisfactory compromises. Natural speech is aided by linguistics since it streamlines the communication process. Objects' shapes, colors, and sizes can be identified with the use of machine learning. All of these systems qualify as expert systems (Eisenhardt and Graebner, 2007).

To address the limitations of the initial computations of standard 2-DEA, the outer variant DEA combines DEA with additional methods including bootstrapping, OLS, and NNs. The use of two-stage DEA has sparked debate. In addition, (Omrany et al., 2016) emphasizes that DEA is unable to deal with the problem of robust data that have fewer required features due to the lack of prior parameter estimation. Multiple studies may make use of model 2's flexibility by including additional variants in line with their research aims, such as using OLS to evaluate the processes governing the connection between factor elements or using computerized approaches.

4.2. Regression analysis

There are a number of reasons why DEA and related techniques are so commonly used in the banking industry. A bank's level is analogous to a company's ability to convert assets into revenues, allowing successful banks to boost profits while decreasing costs and thus stimulating economic growth (Chan and Chan, 2020). Finally, there is a persistent risk that banks must face as they try to improve the accuracy of performance ratings. Since the DEA model has been modified to include components such as loose consequences and fuzzy control through a number of theoretical experiments, it is crucial to examine the efficiency of banks in a consistent and careful manner when constructing models in order to guarantee approximation of effectiveness.

There are often a large number of areas of DEA application in financial services investigations. Financial services were found to be the most frequently examined issue in the comprehensive assessment of the use of DEA-based models for assessing capital adequacy. We conclude that the increased availability of account credentials after the 2008 economic crisis is to blame for the persistent overrepresentation of DEA-based research in the financial sector since 2011 (Endo et al., 2019). The conventional DEA model is also being updated to strengthen its role as a cornerstone of finance research.

By discarding varying quantities of underlying interfaces, for instance, we may directly address slacks instead of relying on the proportional variance across interfaces. Reasons for inefficiency are explained by non-zero margins. The flannel pants DEA method needs both unit inversion and monotonous repetition. Several studies have suggested a superior flannel pants technique for evaluating capital adequacy because of the way flannel pants DEA deals with simultaneous non-proportional modification of interfaces. Traditional DEA can be

created using the tuxedo pants method or the directed distance measure (Nair et al., 2010). The bidirectional distance measure proposed for multi-input and multi-output systems allows both input reductions and output maximization. Thus, it demonstrates the connection between company effectiveness and a directed communication network. The directed proximity metric is often used in banking and finance research to assess efficiency and effectiveness. In particular, we propose a new implementation of the spectrum positional distance framework (RDM), which takes into account the role that achievement plays in the economic organization effectiveness and uses DEA prediction of the positional path length at night before going to sleep. This length is estimated by RDM, and the explanation of the DEA prediction is given for its impact on the treasury efficiency. (Asadi et al., 2014) analyzed the effects of aggregate production and supply and outcome changes on overall bank profitability using a Russell-directed distance model.

Indeed, the Russell scale is a frequent DEA variant. This metric seeks to account for the fact that a lesser quantity of a given input could lead to more waste, as stated by, who devised the metric to contest claim of operational performance (Rouwette et al., 2002). The banking industry and banking performance are evaluated using the Russell scale, which is a non-radial compression of sources. Problems with the poorly formed ideal solution of the DEA model associated with dress pants were addressed by using the normalized Ross measure. Non-performing loans (NPLs) and other unsatisfactory outputs were analyzed by Barros and coworkers using the Roger's dimensional spacing. To analyze revenue inefficiency using a more all-encompassing method that takes into account modern adjustments to inputs/outputs and lagged effects, Russell's assessment is modified to include the unidirectional technique and the flannel pants test.

The prediction accuracy of conventional DEA has been enhanced by other DEA-based hybrid techniques. To find the most efficient financial institutions, Plot-created multi-DEA frameworks in which the model takes a number of different variables and produces a number of different results (H. Liu et al., 2022). The greatest results are achieved within the fuzzy-DEA (FDEA) framework developed by.

"Operative systems" (ABM) are computer simulations that depict individuals or organizations as autonomous entities that interact with each other and their environment. Because ABMs involve goal-directed effort, they may be useful in banking reform, investing, and other monetary activities. Agent-based models consist of four main components: the agent, the topologies, and the environments (Bao et al., 2020). An autonomous agent is one that can think for itself and act in its own best interest. Two ecological elements, the credit crisis, are built into the architecture and represent a potential systemic threat to the whole system.

This makes ABM a useful model for recreating intricate processes in the real world. To be flexible, ABM can be used to capture emergent phenomena and the system can be defined organically. ABM has the potential to recreate the behavior of a system that relies more on the interdependencies between its constituent parts than on their isolated characteristics (Wang et al., 2022). Recently, ABM has been used to mimic bank runs caused by consumers simultaneously choosing to withdraw. It is widely believed by those who were spared the impact of the negative shock that financial institutions will go bankrupt if their customers delay withdrawals until a later date.

Other advantages of ABM, such as easy explanation and adaptability, are bolstered by its ability to simulate impromptu action (Mehrjerdi et al., 2019). This accurate depiction exemplifies the ability of ABM to describe systems by focusing on the actions of discrete, homogenous units rather than on processes occurring at the system level as a whole. Using ABM, the impact of various regulations that affect the monetary base's cash position in diverse financial systems was examined. They suggest that diversity among banks should be portrayed, given that the banking system includes a wide range of firms with varying degrees of management aptitude. ABM makes it possible to experiment with a much larger set of variables by simulating the actions of several agents.

Table 1
Variables description and data sources.

Variables	Description	Data sources
BBG	Building back greener	The China Statistical Yearbook
NR	Natural resource	The China Science and Technology Statistical Yearbook
SD	Sustainable development	The China Industrial Statistical Yearbook
SD2	Quadratic of the Sustainable development	The China Industrial Statistical Yearbook
inf	Infrastructure	The China Labor Statistical Yearbook
hum	Human capital development	The China Labor Statistical Yearbook
market	The market level	The province – level statistical yearbooks
open	The degree of openness	The China Labor Statistical Yearbook
ER	Environmental regulation	The province – level statistical yearbooks

Table 2
Description of variables.

Variables	Obs	Mean	Std. dev.	Min	Max
BBG	418	1.5161	0.5721	0.6097	4.9791
NR	418	7.7390	4.0729	4.7821	14.1750
SD	418	1.3160	0.6071	– 0.4597	4.7977
SD2	418	4.0941	1.5871	0.0000	5.8329
inf	418	4.5959	0.3690	1.3997	3.2660
hum	418	1.7690	0.1797	1.3544	4.3990
market	418	1.8397	0.3180	0.8491	4.4607
open	418	0.2988	0.3591	0.0141	1.7221
ER	418	– 5.7321	0.2861	– 6.6514	– 5.0629

The capacity of ABM to incorporate new models into existing ones exemplifies this adaptability.

To model an approach that improves and integrates its variety, several studies have turned to ABM (Table 12) to examine pandemic threat (H (see Table 1). Liu et al., 2022). The possibility of a domino effect affecting all banks adds to the overall level of systemic risk in the market where a failure in one region might lead to a downturn. Calibration or constructed types of ABM, such as the BH model or actor-based model, have been used in studies to define banking behavior and compare it to ABM results (see Table 3) (see Table 4) (see Table 5)

Table 3
Impact of SD on BBG: benchmark regression results.

Variables	(1) FE	(2) Difference GMM	(3) System GMM	(4) FE	(5) Difference GMM	(6) System GMM
L.BBG		1.1600 ** (17.4519)	1.0590 *** (31.1314)		0.9797 *** (17.2249)	0.9541 *** (17.7019)
NR	0.1544 (1.6317)	0.1946 ** (6.1150)	0.0976** (3.2588)	– 0.3019 (– 1.6313)	– 0.5275 *** (– 4.8741)	– 0.6948 *** (– 5.6790)
NR2				0.0122 (1.3280)	0.0539 *** (7.6952)	0.0531 *** (13.2951)
SD	– 0.4431 (– 1.3492)	– 1.0214 * ** (– 5.1959)	– 0.5521 *** (– 4.0722)	– 0.3615 (– 1.0088)	– 0.1690 (– 0.5964)	– 1.5407 *** (– 4.0388)
SD2	0.2997 * * (3.1688)	0.0771 (1.1840)	0.1390* (1.8787)	0.31 (1.7690)	0.39 * ** (7.8387)	0.2818 * ** (4.4549)
inf	0.0060 (0.2915)	0.0490 *** (5.8381)	0.0078 *** (5.4588)	0.0295 (1.6170)	0.0244 *** (3.1007)	0.0147 * ** (3.1807)
hum	0.3131 (0.4380)	0.8773 *** (3.3714)	0.1780 (1.0180)	0.0261 (0.0459)	0.9661 *** (3.4005)	0.7972 * ** (4.0660)
market	– 0.0041 (– 0.0190)	– 0.2774 * ** (– 3.7599)	– 0.0688 (– 0.8776)	– 0.2241 (– 0.6840)	– 0.4961 *** (– 6.8931)	– 0.2402 * ** (– 3.0616)
open	0.5151* (1.7840)	0.2312 * * (3.0690)	0.0931 (0.7705)	0.3366 (1.0431)	0.5818 *** (6.8570)	0.06880 (0.6441)
ER	– 0.0741 (– 0.7604)	– 0.0677 * * (– 3.4660)	– 0.0588 * * (– 3.4551)	– 0.0701 (– 0.7631)	– 0.1441 * ** (– 4.7710)	– 0.0688 * * (– 3.5197)
.cons	– 0.8141 (– 0.4888)	– 1.4681 * ** (– 4.8750)	– 0.58970 * ** (– 5.6951)	0.9441 (0.6466)	0.6581 (1.3661)	0.9799* (1.6717)
AR(1)		0.0006	0.0007		0.0007	0.0004
AR(2)		0.6951	0.6507		0.750	0.5521
Sargan testP – value		1.0000	1.0000		0.9031	1.0000
Individual fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
N	418	362	393	422	362	393
R2	0.621			0.736		

(see Table 6) (see Table 7) (see Table 8) (see Table 9) (see Table 10) (see Table 2).

Only 1759 positions were open in Guangdong in a particular month, which is about 60% of the maximum number of positions accessible in other areas. The epidemic has had little effect on it so far. While both, the former is more affected by the concentration of business web offices in Shenzhen. Compared to the other three cities (Han et al., 2022). The graduation season in major Chinese cities such as Nanjing and Guangzhou in July causes a surge in demand for jobs in the following months. Beijing has not seen an increase in the number of job seekers since the local outbreak of the pandemic in June, when preventive measures and warning signs were bolstered (H. Liu, Wu et al., 2022). The pandemic has had a negative impact, as seen by the decline in employment competitiveness. Testing and inspection jobs declined by 43%, while electronic application growth increased by approximately 25%. Due to the rapid advancement in artificially acquired knowledge, this pandemic has had far-reaching effects on the post-structure and has intensified polarization in the labor market. When the high and medium zones in Beijing were 82, the demand for occupations requiring artificial intelligence increased by 65%. Table 11 shows Hebei province as the epidemic's nerve center, which has far-reaching effects on Beijing's labor force. The data suggest that the intensity of the pandemic is inversely related to the number of jobs in the field of AI technology.

Table 4

The impact of sustainable development on building back greener: regression results of the LSDVC method.

Variables	(1)	(2)	(3)	(4)
	Difference GMM	System GMM	Difference GMM	System GMM
<i>L.BBG</i>	1.0701 *** (41.6541)	1.1088 *** (70.660)	1.0896 *** (50.0790)	1.0907 *** (59.3541)
<i>NR</i>	0.1371 *** (3.1880)	0.0750* (1.9688)	− 0.1177* (− 1.7614)	− 0.1270 ** (− 3.3664)
<i>NR2</i>			0.0088* (1.9690)	0.0094 *** (2.5994)
Control variables	Yes	Yes	Yes	Yes
<i>N</i>	388	388	388	388

Table 5

Regression findings for the replacement of explanatory variables for sustainable development the market on building back greener.

Variables	(1)	(2)	(3)	(4)
	Difference GMM	System GMM	Difference GMM	System GMM
<i>L.gtfp</i>	1.2129 *** (22.2961)	1.0770 *** (41.2029)	0.6410 *** (3.4929)	1.0529 *** (14.2883)
<i>aie</i>	0.2866 *** (7.6339)	0.0873 *** (6.0004)	− 1.2339* (− 1.7350)	− 0.4101* (− 1.9491)
<i>aie2</i>			0.0880 ** (3.0750)	0.0260 ** (3.2077)
<i>_cons</i>	− 4.1131 ** (− 3.2503)	− 1.2066 ** (− 3.4914)	4.0751 (1.6349)	0.4995 (0.4888)
Control variables	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0041	0.0021	0.0006	0.0031
<i>AR(2)</i>	0.8990	0.5097	0.7460	0.5280
Sargan testP – value	1.0000	1.0000	1.0000	1.0000
<i>N</i>	239	266	239	266

Table 6

Data from other regions in the same area were used as an instrumental variable in the regression to estimate the effects of artificial intelligence on building back greener.

Variables	(1)	(2)	(3)	(4)
	Difference GMM	System GMM	Difference GMM	System GMM
<i>L.BBG</i>	1.1281 *** (18.9144)	1.0681 *** (31.6214)	0.9780 *** (7.8131)	0.9941 *** (17.6421)
<i>NR</i>	0.2181 *** (5.5621)	0.0441 *** (3.8770)	− 0.4115* (− 1.9667)	− 0.3703 *** (− 4.0860)
<i>NR2</i>			0.0361 *** (3.0590)	0.0250 *** (3.4450)
<i>_cons</i>	− 1.4439 *** (− 4.7550)	− 0.4406** (− 2.1361)	− 0.1304 (− 0.3500)	0.3841* (1.6830)
Control variables	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0021	0.0006	0.0041	0.0007
<i>AR(2)</i>	0.6931	0.6504	0.5990	0.5401
Sargan testP – value	1.0000	1.0000	1.0000	1.0000
<i>N</i>	359	388	359	388

Sustainable development has helped in areas such as healthcare and workforce development. Using the data, we were able to infer two scenarios where SD has a wide influence on research and development. According to Conjuring up AI, the annual rate of increase in SD-related patent applications went from 40,000 to 70,000 between 2004 and 2018. Furthermore, 42% of our entire class time was spent on automation related materials. The centrality of humanity and the pervasiveness of human beings in our civilization can both be observed here.

5. Conclusion and policy implications

This study offers vital insights into the effectiveness of policy interventions in China's natural resource markets from 2000 to 2020. This era witnessed significant economic expansion alongside environmental degradation in the country. By employing a Panel Data model, this study

successfully captured the longitudinal effects of diverse policy interventions, providing a more thorough understanding than previous research endeavors.

The findings unequivocally show that the optimal strategy, encompassing regulatory mechanisms and market-driven incentives, is the most productive in fostering sustainable practices in natural resource markets. The study found that more is needed than relying on regulatory measures alone, emphasizing the necessity of adopting a comprehensive approach that includes market incentives to promote sustainable practices among producers and consumers. This is of particular importance for China, considering its substantial contribution to global resource use and the potential implications for global sustainability endeavors.

Resource efficiency refers to the optimal use of resources to minimize waste and maximize output. The findings of the Panel Data model indicate a statistically significant relationship between market-based

Table 7

Regression findings using optical cable density as an instrumental variable demonstrate the effect of artificial Intelligence on green economic growth.

Variables	(1)	(2)	(3)	(4)
	<i>Difference GMM</i>	<i>System GMM</i>	<i>Difference GMM</i>	<i>System GMM</i>
<i>L.BBG</i>	1.1364 *** (17.6688)	1.0490 *** (31.3449)	1.0195 *** (12.7544)	0.9900 *** (21.5566)
<i>NR</i>	0.2718 *** (6.2877)	0.0788 ** (3.2629)	− 0.5459 *** (− 3.9394)	− 0.3531 *** (− 5.8881)
<i>NR2</i>			0.0450 *** (4.0480)	0.0250 *** (6.4929)
<i>_cons</i>	− 1.4702 *** (− 4.8588)	− 0.5877 *** (− 3.6850)	0.5821 (1.1918)	0.3590* (1.7590)
<i>Control variables</i>	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0021	0.0006	0.0039	0.0006
<i>AR(2)</i>	0.7081	0.6490	0.6429	0.5859
<i>Sargan test p – value</i>	1.0000	1.0000	1.0000	1.0000
<i>N</i>	359	388	359	388

Table 8

Impact of SD on BBG: regression results of DID.

Variables	(1)	(2)	(3)	(4)
	<i>DID</i>	<i>Placebo testSham experimental group</i>	<i>Placebo testSham shock time</i>	<i>The parallel trend test</i>
<i>Treat_Post</i>	0.1614 *** (3.6811)	− 0.2431 (− 0.3480)	0.0751 (1.2990)	
<i>pre_3</i>				0.0788 (0.9277)
<i>pre_2</i>				0.0660 (0.6177)
<i>Current</i>				0.0197 (0.1959)
<i>post_1</i>				0.0451 (0.4541)
<i>post_2</i>				0.1929* (1.6944)
<i>post_3</i>				0.2688 ** (3.3961)
<i>post_4</i>				0.3859 *** (4.6184)
<i>post_5</i>				0.6749 *** (4.9259)
<i>Control variables</i>	Yes	Yes	Yes	Yes
<i>Individual fixed effect</i>	Yes	Yes	Yes	Yes
<i>Year fixed effects</i>	Yes	Yes	Yes	Yes
<i>N</i>	418	213	418	418
<i>R2</i>	0.741	0.714	0.741	0.770

Table 9

Impact of SD on BBG: regression results for resource-rich areas.

Variables	(1)	(2)	(3)	(4)
	<i>Difference GMM</i>	<i>System GMM</i>	<i>Difference GMM</i>	<i>System GMM</i>
<i>L.BBG</i>	1.2541 *** (29.8941)	1.1470 *** (31.1818)	1.0510 *** (14.7570)	1.0700 *** (29.1373)
<i>NR</i>	− 0.0590 *** (− 4.9597)	− 0.0198 ** (− 1.9710)	− 0.7950 ** (− 3.1580)	− 0.0870 *** (− 4.0790)
<i>NR2</i>			0.0612 ** (4.3721)	0.0088 *** (3.1081)
<i>_cons</i>	− 0.9494 *** (− 4.0780)	0.0690 (0.3571)	− 0.2680 (− 0.6170)	− 0.1870 (− 0.8670)
<i>Control variables</i>	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0013	0.0007	0.0007	0.0005
<i>AR(2)</i>	0.6554	0.6629	0.6729	0.8521
<i>Sargan testP – value</i>	1.0000	1.0000	1.0000	1.0000
<i>N</i>	359	388	359	388

incentives and enhanced resource efficiency among producers. It is recommended that policymakers carefully consider the implementation of tax incentives or subsidies to encourage enterprises to adopt resource-

efficient technologies. It has been noted that the impact of regulatory actions, such as the implementation of more stringent environmental standards, on consumer behavior is good, as it promotes consumers'

Table 10
Impact of SD on BBG: regression results for the resource-general areas.

Variables	(1)	(2)	(3)	(4)
	Difference GMM	System GMM	Difference GMM	System GMM
<i>L.BBG</i>	1.2466 *** (21.5248)	3.0924 *** (29.1248)	3.0606 *** (13.8559)	3.0977 *** (42.4860)
<i>NR_g</i>	0.1274 ** (3.9839)	0.0170* (3.7945)	− 0.7492 *** (− 4.6871)	− 0.0321 ** (− 4.1440)
<i>NR_g2</i>			0.0519 *** (4.2157)	0.0059 *** (4.8131)
<i>_cons</i>	− 3.4531 *** (− 3.6949)	− 0.4213 (− 4.131)	0.3562 (3.0777)	− 0.4588 ** (− 4.1131)
<i>Control variables</i>	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0007	0.0007	0.0014	0.0005
<i>AR(2)</i>	0.5197	0.52131	0.5470	0.5371
<i>Sargan test P – value</i>	1.0000	1.0000	1.0000	1.0000
<i>N</i>	359	388	359	388

Table 11
Impact of SD on GTC index.

Variables	(1)	(2)	(3)	(4)
	Difference GMM	System GMM	Difference GMM	System GMM
<i>L.GTC</i>	2.2838 *** (60.1231)	2.2352 *** (64.4650)	2.2659 *** (31.6396)	2.1807 *** (29.8960)
<i>NR</i>	0.0669 *** (3.9360)	0.0471 *** (3.0088)	− 0.2244* (− 3.7771)	− 0.1806 ** (− 2.9604)
<i>NR2</i>			0.0170 *** (4.7519)	0.0158 *** (3.1231)
<i>_cons</i>	− 0.3918 (− 3.0031)	− 0.3896 (− 3.3247)	0.3215 (0.6315)	0.1684 (0.3880)
<i>Control variables</i>	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0000	0.0000	0.0000	0.0000
<i>AR(2)</i>	0.8388	0.9812	0.9641	0.8241
<i>Sargan testP – value</i>	1.0000	1.0000	1.0000	1.0000
<i>N</i>	359	388	359	388

Table 12
Impact of SD on GEC index.

Variables	(1)	(2)	(3)	(4)
	Difference GMM	System GMM	Difference GMM	System GMM
<i>L.GEC</i>	0.4797 *** (5.6741)	0.6321 *** (6.9418)	0.0959 (3.1315)	0.7159 *** (7.6931)
<i>NR</i>	− 0.0243 *** (− 4.7400)	− 0.0277 ** (− 4.2114)	− 0.0688 ** (− 4.3590)	− 0.1730 ** (− 4.1481)
<i>NR2</i>			0.0071 *** (4.9181)	0.0084* (3.7341)
<i>_cons</i>	0.4770 ** (4.0271)	0.2371 (0.8270)	0.4351 *** (3.7431)	0.8641 ** (3.0806)
<i>Control variables</i>	Yes	Yes	Yes	Yes
<i>AR(1)</i>	0.0021	0.0014	0.0007	0.0048
<i>AR(2)</i>	0.4570	0.5266	0.3302	0.8221
<i>Sargan test P – value</i>	0.9997	1.0000	0.9841	1.0000
<i>N</i>	359	388	359	388

inclination to purchase sustainable items. In terms of policy implications, a combination of public awareness campaigns and regulatory measures has the potential to effectively influence consumer behavior towards sustainable choices. Research has revealed that geographical areas that engage in active public-private partnerships demonstrate superior sustainable outcomes in their natural resource markets.

Facilitating partnerships between the public and commercial sectors has the potential to accelerate the adoption of sustainable practices. The effectiveness of local interventions exhibited heterogeneity, but national policies demonstrated a more consistent influence. It is recommended to

establish a comprehensive policy framework at the national level to promote uniformity in the implementation of sustainable practices in different regions. The findings from the Panel Data model suggest that the beneficial outcomes of balanced policy measures are not immediately apparent but manifest themselves more prominently over an extended period. Regarding policy implications, policymakers should anticipate the necessity of a sustained and enduring dedication to guarantee the triumph of sustainable development endeavors.

Advances in technology present new avenues for the pursuit of sustainable development. Subsequent investigations should examine the

ramifications of nascent technology on natural resource markets and the effectiveness of existing policies. By acknowledging and overcoming these constraints and adhering to the proposed avenues for future investigation, we can advance the results of the current study and cultivate more effective and comprehensive approaches to sustainable development.

The analysis was constrained by data limitations, namely data availability and quality, notably during the initial years of the 2000–2020 period. Further investigation could benefit from the implementation of more rigorous data collection methodologies. The present analysis focused on comprehensively examining specific sectors within the natural resource markets. Subsequent investigations can focus on specific industries such as mining, forestry, or water resources to yield more intricate and comprehensive understandings. The study failed to consider cultural aspects that could influence the effectiveness of policy interventions. Further investigation is warranted to examine the potential influence of cultural views on policy outcomes in the context of sustainability.

Author statement

Xi Chen; Writing - Original Draft, Writing - Review & Editing, Investigation.

Binyi Liu Methodology, Validation, Project administration, Funding acquisition.

Yali Ma; Formal analysis, Investigation.

Shuhui Lv; Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.resourpol.2023.104220>.

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